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Review article

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Seasonal CO_2 amplitude in northern high latitudes

Zhihua Liu[1,2,29], Brendan M. Rogers[3,29], Gretchen Keppel-Aleks[4,29], Manuel Helbig[5,6,29], Ashley P. Ballantyne[7,8,29], John S. Kimball[1,29], Abhishek Chatterjee[9], Adrianna Foster[10], Aleya Kaushik[11], Anna-Maria Virkkala[3], Arden L. Burrell[3], Christopher Schwalm[3], Colm Sweeney[12], Edward A. G. Schuur[13], Jacqueline Dean[3], Jennifer D. Watts[3], Jinhyuk E. Kim[14], Jonathan A. Wang[15], Lei Hu[12], Lisa Welp[16], Logan T. Berner[17], Marguerite Mauritz[18], Michelle Mack[19], Nicholas C. Parazoo[9], Nima Madani[20], Ralph Keeling[21], Roisin Commane[22], Scott Goetz[17], Shilong Piao[23], Susan M. Natali[3], Wenjuan Wang[24], Wolfgang Buermann[25], Xanthe Walker[19], Xin Lin[8], Xuhui Wang[23], Yuming Jin[21], Kailiang Yu[26] & Yangjian Zhang[27,28]

Abstract

Global climate change is influencing the seasonal cycle amplitude of atmospheric CO_2 (SCA), with the strongest increases at northern high latitudes (NHL; >45° N). In this Review, we explore the changes and underlying mechanisms influencing the NHL SCA, focusing on Arctic and boreal terrestrial ecosystems. Latitudinal gradients in the SCA are largely governed by seasonality in temperature and primary production, and their influence on ecosystem carbon dynamics. In the NHL, the SCA has increased by 50% since the 1960s, mostly due to enhanced seasonality in net carbon dioxide (CO_2) exchange in NHL terrestrial ecosystems. Temperature most strongly influences this trend, owing to warming impacts on growing season length and plant productivity; CO_2 fertilization effects have a secondary role. Eurasian boreal ecosystems exert the strongest influence on the SCA, and spring and summer are the most influential seasons. Enhanced ecosystem respiration during the non-growing season exhibits most uncertainty in the SCA response to global and landscape drivers. Observed changes in the seasonal amplitude are projected to continue. Key priorities include extending carbon flux and ecosystem observation networks, particularly in tundra ecosystems, and including drivers such as vegetation cover and permafrost in process models to better simulate seasonal dynamics of net CO_2 exchange in the NHL.

Sections

A full list of affiliations appears at the end of the paper. e-mail: zhangyj@hbu.edu.cn

Introduction

Anthropogenic greenhouse gas emissions have strongly altered the global carbon cycle. For instance, atmospheric carbon dioxide (CO_2) concentrations increased from 280 ppm during the 1860s pre-industrial era to 420 ppm in 2023 (refs. 1–3). In parallel, seasonal changes in atmospheric CO_2 have also been observed[4,5]. These changes are reflected in the seasonal cycle amplitude of atmospheric CO_2 (SCA) – the difference between the maximum annual atmospheric CO_2 (representing net ecosystem respiration and CO_2 release) and the minimum annual atmospheric CO_2 (net ecosystem photosynthesis and CO_2 uptake) – at a given location[5,6] (Box 1). Thus, the SCA provides insight into large-scale variations in the terrestrial carbon balance that are otherwise not easily measured by localized measurements, such as eddy covariance towers or field studies, which are limited in spatial coverage and do not necessarily capture broader, regional-scale processes[7–9].

The SCA has generally increased globally. Compared with the tropics and lower latitudes, these increases are particularly strong in the northern high latitudes (NHL; >45° N). Here, the SCA in the lower troposphere increased by ~50% between the 1960s and 2010s (refs. 5,6,8), reflecting enhanced productivity in the growing season[8,10] and greater respiration in shoulder and non-growing seasons[11–13]. However, understanding of the underlying drivers of these changes is limited: regional patterns and environmental drivers of gross primary productivity, respiration and fluxes from disturbance and lateral transfer are poorly constrained; and contributions from changes in vegetation cover, disturbance regimes and permafrost thaw lack observational evidence[14,15]. Accordingly, the accuracy of process models and climate projections is limited, as evidenced by an underestimation of the modelled SCA and competing evidence of dominant drivers[16–18].

As such, there is a pressing need to better constrain the drivers governing SCA trends. Understanding these drivers, including their regional patterns, responses to climate change and interactions, is key to constraining historic and current carbon–climate feedbacks in high latitudes and supporting climate policy with process-level insights. The importance of these feedbacks will likely grow in the future; for example, the carbon feedback attributed to permafrost thaw and degradation is estimated to be equivalent to that of the global economy during the remainder of the twenty-first century[19]. To inform robust and accurate modelling under policy-relevant climate scenarios and meet climate targets, these emissions and SCA trends need to be adequately accounted for to ensure estimates of remaining carbon budgets are accurate.

In this Review, we summarize the terrestrial processes and environmental drivers contributing to SCA trends in the NHL. We begin by presenting the global and hemispheric-scale distributions of the SCA, focusing on temporal and regional patterns in the NHL. We then synthesize and distil knowledge of the seasonal dynamics of CO_2 drawdown and release, regional and biome contributions, and drivers of increasing SCA based on a survey of the literature and expert opinion. We next illustrate the impact of these drivers on observed SCA trends at the Barrow Atmospheric Baseline Observatory in Utqiaġvik, Alaska, USA (Point Barrow). Finally, we put forward recommendations to improve monitoring networks and simulations of seasonal carbon dynamics.

Enhanced seasonal amplitude

In the NHL, seasonality in CO_2 concentrations consists of a peak in late spring and a trough in late summer. The peak follows sustained decomposition of soil organic matter over winter and shoulder season months, just prior to widespread activation of photosynthesis in spring, and the trough follows the growing season during which ecosystem photosynthesis exceeds respiration, resulting in a drawdown of CO_2 (ref. 20). Here, the SCA is considered as the peak-to-trough difference (SCA_{P-T}) (see Box 1), and the observed and projected future latitudinal and temporal trends are discussed.

Latitudinal and temporal trends

Globally, there is a strong latitudinal gradient in the mean SCA. The SCA is weaker with later phasing at lower latitudes compared with higher latitudes. In the southern hemisphere, a much weaker cycle is found (Fig. 1a), owing to the smaller land area and more temperate climates resulting in diminished seasonality of temperature and plant activity[1]. For example, long-term in situ measurements have shown that SCA_{P-T} is between 15 and 20 ppm at Point Barrow (71° N), and diminishes southwards to about 5 ppm near the Equator (Mauna Loa, 20° N). Atmospheric inverse modelling can estimate these large-scale latitudinal features of the SCA between sparse surface sampling observatories. Through combining atmospheric transport models with observed CO_2 gradient mole fractions, such models can provide quantitative estimates of surface flux variations over time. For example, atmospheric inversion modelling from the Copernicus Atmospheric Modelling Service (CAMS) framework showed that the latitudinal mean SCA increases from less than 2.5 ppm in the southern hemisphere to 5–7.5 ppm in the northern mid-latitudes, to ~10 ppm in the NHL (Fig. 1a).

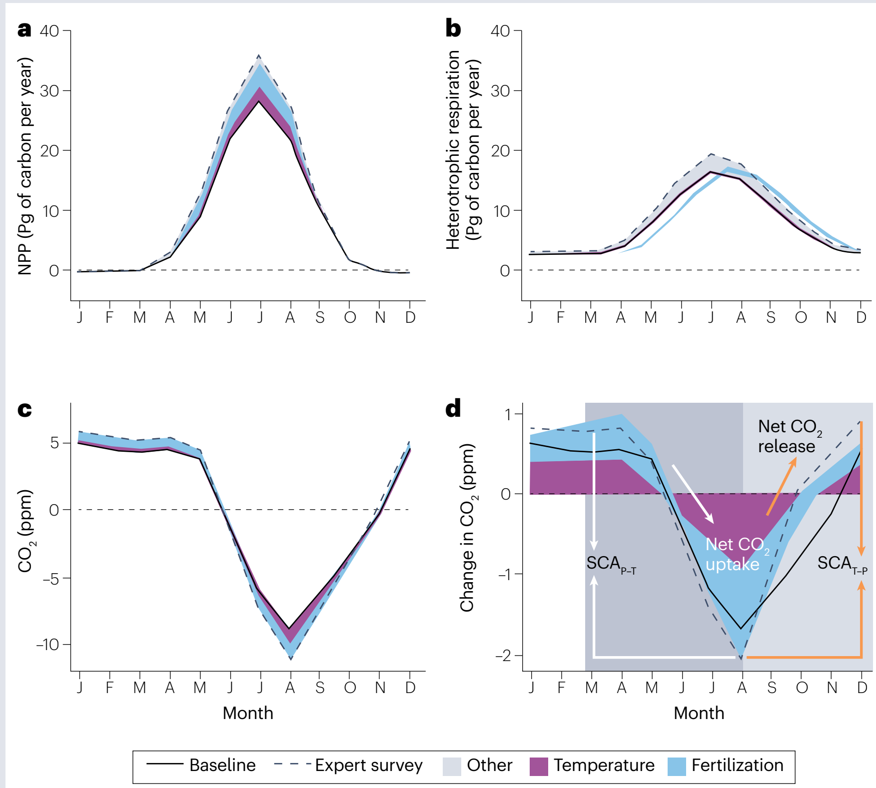
The global network of atmospheric CO_2 monitoring stations has allowed long-term assessment of the SCA at the hemispheric scale since the 1960s and at regional scales since the 1980s. Long-term increases in the SCA were first noted in the 1980s at Point Barrow[21,22], but later analyses of longer time series with expanded spatial coverage show the SCA increasing at the global scale[23]. However, SCA trends have been persistently much more pronounced in the NHL than in other regions since the 1960s (refs. 5,6,8).

Since the 1980s, the temporal trend of SCA_{P-T} has generally increased with latitude in the northern hemisphere (see Supplementary Table 1). Zonally averaged SCA_{P-T} increased from 0.032 ± 0.036 ppm per year at 0–45° N to 0.043 ± 0.145 ppm per year in the NHL based on in situ CO_2 measurements, and the SCA_{P-T} derived from CAMS inverse modelling showed the same pattern with the zonally averaged SCA_{P-T} increasing from 0.004 ± 0.031 ppm per year in the 0–45° N latitude zone to 0.046 ± 0.010 ppm per year in the NHL (Fig. 2). The more rapid increase in SCA_{P-T} in the NHL compared with low latitudes and the southern hemisphere implies more dramatic changes in terrestrial carbon cycle processes in the NHL and, potentially, stronger climate–carbon feedback that merit a better understanding of the underlying mechanisms. Indeed, observations indicate that the increases in the NHL SCA are primarily driven by heightened seasonality in net CO_2 exchange by terrestrial ecosystems, with minor contributions from other processes or sources[8,24,25].

Increasing SCA trends could be due to decreasing trends of minimum atmospheric CO_2 (trough), suggesting more net CO_2 uptake, or increasing trends of maximum atmospheric CO_2 (peak), suggesting more net CO_2 release, or a combination of the two processes. CAMS inversion modelling indicates that the trough value decreased substantially in the extra-tropical northern hemisphere, and that the trend is much larger in the NHL (−0.043 ± 0.007 ppm per year for 45° N–90° N) than in the mid-low latitudes (−0.0055 ± 0.018 ppm per year for 0° N–45° N) (Fig. 3). This latitudinal difference suggests an increase in net CO_2 uptake from plant photosynthesis with latitude during the growing season (Fig. 3). The decrease in tough values is of greater magnitude than the increase in peak values, implying that enhanced growing season productivity is the major process contributing to increasing the SCA

Box 1 | Estimating and modelling SCA

The seasonal cycle amplitude of atmospheric CO_2 (SCA) refers to the difference between maximum (peak) and minimum (trough) values of the detrended seasonal cycle of atmospheric carbon dioxide (CO_2) for each calendar year. The SCA is governed by the magnitude of seasonal cycles in net primary production (NPP) (see the figure, panel **a**) and heterotrophic respiration (see the figure, panel **b**) and their influence on atmospheric CO_2 (panel **c**). The SCA can be calculated as the peak-to-trough difference ($SCA_{P\text{-}T}$), representing cumulative net CO_2 uptake during the growing season, or as the trough-to-peak difference ($SCA_{T\text{-}P}$), representing cumulative net CO_2 release due to respiration during the non-growing season; the magnitude and trends of each are similar[6] (see the figure, panel **d**). In situ-based $SCA_{P\text{-}T}$ was derived from weekly CO_2 data from 26 atmospheric CO_2 monitoring stations with at least 25 years between the 1980s and 2020 (ref. 177). $SCA_{P\text{-}T}$ derived from inversion data applied an optimized column-mean dry air mole fraction of CO_2 at 1.9° x 3.75° spatial resolution between 1980 and 2020 (ref. 178).



The impact of different drivers on the observed SCA signal was estimated and the change in CO_2 amplification was quantified using a pulse-response transport operator derived from an atmospheric transport model[180], informed by expert opinion on the trends in component fluxes. Shaded areas in all panels in the figure represent the imprint of expert-estimated trends in temperature-driven and CO_2 fertilization-driven changes to NPP and heterotrophic respiration, and resultant effects on SCA (see the figure, panel **d**). At the Barrow Atmospheric Baseline Observatory in Utqiaġvik, Alaska, USA (Point Barrow), temperature-driven trends resulted in a 1.4 ppm increase in the SCA, mostly from temperature impacts on NPP (1.0 ppm). CO_2 fertilization also increased the amplitude of CO_2 by 1.4 ppm. Collectively, other drivers accounted for only a 0.1 ppm increase in the SCA. Therefore, most of the simulated increases of the seasonality of CO_2 at Point Barrow are explained by changes in NPP and heterotrophic respiration caused by temperature and CO_2 fertilization, highlighting the dominant role of the terrestrial carbon cycle in shaping the NHL SCA. However, the total change in heterotrophic respiration is governed by a complex set of drivers including permafrost thaw, snowpack dynamics and wildfire.

in the NHL. Enhanced growing season productivity is mainly driven by relaxation of temperature constraints and CO_2 fertilization in the NHL.

Non-growing season respiration is also emerging as an increasingly important driver enhancing the SCA trend. The weaker increasing trend in the SCA peak over the NHL (0.0026 ± 0.0054 ppm per year for 45° N–90° N) suggests an increase in net CO_2 release from respiration during the non-growing season (Fig. 3). In addition, an increasing trend in respiration in autumn and early winter, collectively driven by climate warming, permafrost thaw and a higher availability of decomposable organic matter from previous season productivity, has also been observed[26].

Although trends in the SCA are mainly positive over the past 40 years, they have not been constant[9,23]. Despite a consistent positive trend for NHL sites, the long-term positive trends in the SCA for many low-latitude or southern hemisphere sites are primarily driven by the period after 2000 (ref. 23). CAMS inversion modelling also demonstrated that the SCA trend increased from 1980–2000 to 2001–2020 over about 90% of the northern hemisphere (see Supplementary Figure 1), and for five out of nine in situ long-term NHL measurement stations covering periods from 1980 to 2020 (see Supplementary Table 1).

Future projections

Projections suggest that the SCA will continue to increase during the twenty-first century. The rate of SCA increase is expected to be faster in the NHL than in lower latitudes, similar to historical patterns[27]. Future projections of the SCA from Coupled Model Intercomparison Project Phase 6 (CMIP6) models indicate the SCA will increase by 75% ± 10% by 2081–2100 compared with 1980–2020 under the high-emission climate scenario[28] (Fig. 1b). The magnitude of future SCA increase in the NHL (0.75% per year or 0.09 ppm per year) is similar to historical trends calculated by a network of observations and modelling[5,8]. Although global-scale analyses suggest that the SCA has increased at a faster rate

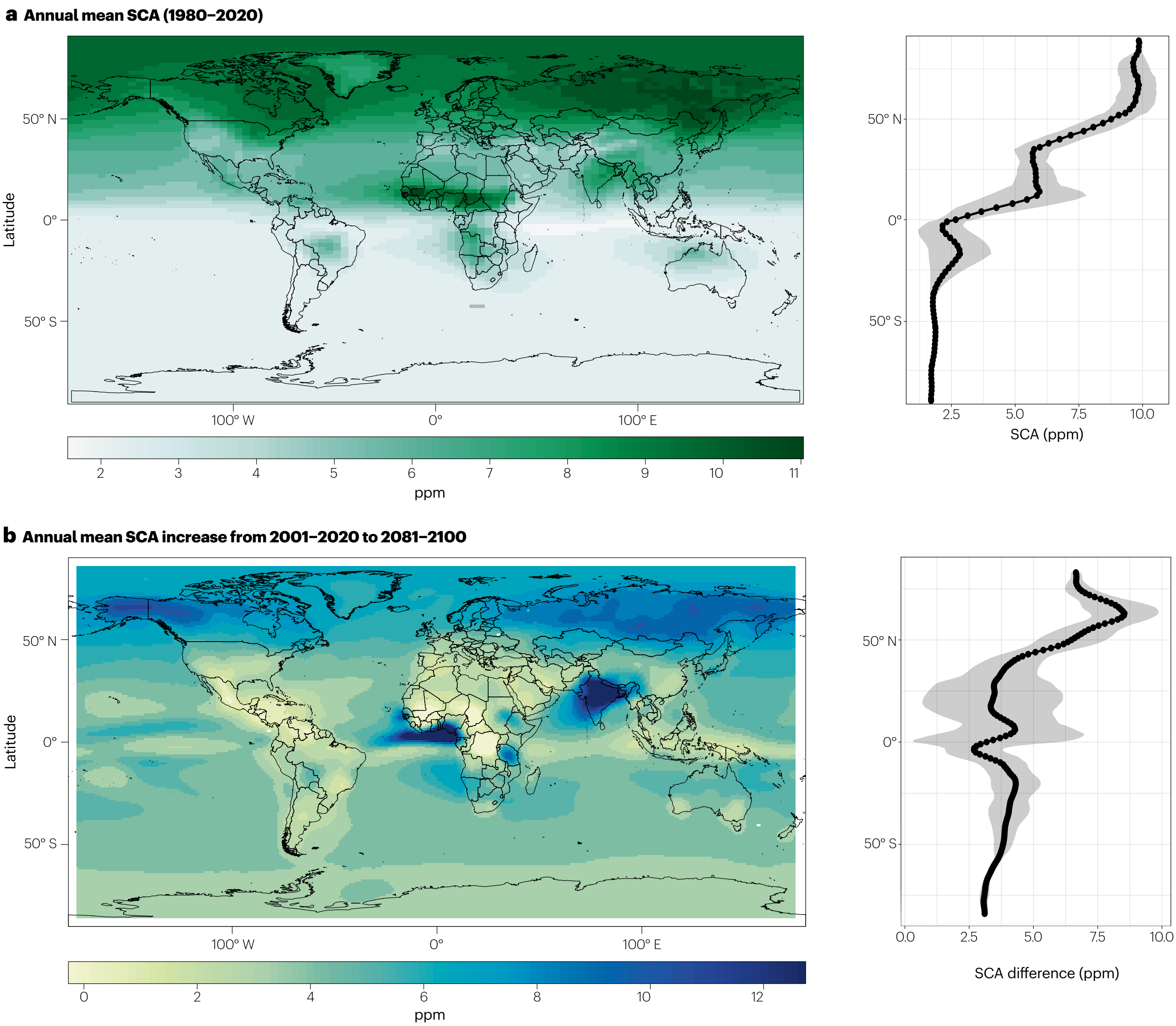


Fig. 1 | Observed and projected $SCA_{P\text{-}T}$. **a**, Gridpoint (left) and zonal average (right) annual mean peak-to-trough seasonal cycle amplitude of atmospheric CO_2 ($SCA_{P\text{-}T}$) during 1980–2020 using inverse modelling data[178]. **b**, As in panel **a**, but for the projected difference in $SCA_{P\text{-}T}$ from 1980–2020 to 2081–2100 based on a five-model ensemble of Coupled Model Intercomparison Project Phase 6 (CMIP6) simulations under SSP5–8.5 (ref. 179). Northern high latitudes (NHL) have larger $SCA_{P\text{-}T}$ in comparison with lower latitudes and the southern hemisphere.

in the NHL regions since 1980 (ref. 23), the cumulative uncertainties in attributing past SCA trends renders these future projections highly uncertain. Therefore, determining the mechanisms and drivers of the enhanced SCA can help diagnose the underlying processes that are key to reducing uncertainties in predicting the high-latitude carbon cycle, and associated climate feedbacks.

Mechanisms of change

Despite uncertainty regarding the underlying mechanisms, some key drivers have been identified. Heightened seasonality in net terrestrial CO_2 exchange, owing to enhanced growing season photosynthesis, accounts for more than 90% of the rise in the SCA in the NHL[8,24,25] (Fig. 4a). The spatially homogeneous increase in SCA trends over the NHL (Fig. 2a) is indicative of broad-scale changes in the carbon cycle owing to changing climate and environmental conditions, rather than other more localized processes. In particular, Arctic and boreal terrestrial ecosystems are the dominant regions contributing to enhanced SCA trends in the NHL[5,8,29]. In contrast, although enhanced air–sea CO_2 flux seasonality due to ocean acidification was found to be a major driver for the increasing SCA since 2000 in the southern hemisphere[23], the overall contribution from ocean CO_2 fluxes to SCA trends in the NHL is very small or negligible[10,23]. Similarly, the contribution of fossil fuel emissions to SCA trends in the NHL is also small or negligible[10,23]. However, other human-related activities in the mid-low latitudes of the northern hemisphere, such as agricultural intensification, do contribute to increasing SCA in the NHL through atmospheric transport[29,30], but their role is uncertain[29]. The influence and contributions of atmospheric transport, seasonal flux dynamics and the biome are now discussed.

Role of atmospheric transport

Atmospheric transport impacts latitudinal and seasonal trends in the SCA. The role of atmospheric transport in shaping the NHL SCA is generally assessed using coarse-resolution atmospheric transport model simulations. These simulations can be forward simulations with assumed fluxes, or they can be inverse simulations, constrained by sparse atmospheric CO_2 measurements. The influence of atmospheric transport, which can advect seasonal flux signatures from regions further south into the NHL, on amplification of the NHL SCA is hard to precisely quantify due to uncertainty among transport models and a lack of independent constraints to evaluate simulated transport[29–31]. Nevertheless, evidence suggests that the major contributor to surface-level SCA increases at high latitudes is enhanced land–atmosphere CO_2 fluxes from boreal and Arctic terrestrial ecosystems[29].

Atmospheric transport can have a substantial impact on the SCA in the troposphere[29,30], with the influence of terrestrial carbon fluxes on the SCA decreasing with altitude[30]. For example, a fixed increase in the seasonality of carbon exchange at 30° N would have a larger impact on the NHL SCA than the same increase in carbon exchange at 60° N due to isentropic transport within the free troposphere[30]. However, the meridional pattern of SCA amplification is not consistent with mid-latitudes being the dominant contributor, as it would result in a smaller north–south gradient in amplification than is observed. In another example, north of 50° N, NHL terrestrial ecosystems contribute nearly 70% of surface NHL SCA trends, but only contribute 55% at the National Oceanic and Atmospheric Administration (NOAA) surface observing sites due to differing regional net CO_2 exchange impacts[29]. Together, this evidence indicates that transported mid-latitude fluxes substantially influence the SCA, and also highlights the dominant (or co-dominant) role of NHL ecosystems in high-latitude SCA amplification[29,30].

However, variability among transport models[31] and uncertainty in parameterized convection[32] lead to substantial uncertainty in quantifications of the role of atmospheric transport on NHL SCA amplification. Even within the NHL, the spatial heterogeneity of ecosystems and their responses to environmental and climatic change lead to spatially heterogeneous patterns of SCA trends[29] (Fig. 2a), complicating the identification of biomes with a dominant influence on the NHL SCA. Given these uncertainties, it is challenging to precisely attribute the NHL SCA amplification to either local NHL impacts or transported mid-latitude impacts. Nevertheless, changes to NHL ecosystems are important for amplification, even assuming generous error bars.

Seasonal flux dynamics and contributions

The key drivers and biome contributions to seasonal dynamics of carbon cycle processes that have driven the change in the NHL SCA are now discussed. This semi-quantitative evaluation of the primary drivers contributing to the increasing SCA in the NHL from 1980 to 2020 is based on a synthesis of the literature and expert opinion (see Supplementary Texts 1 and 2). The relative direct influence of each driver on seasonal carbon flux is taken as the mean value provided by experts, weighted by the confidence levels associated with each flux. Definitions of drivers, seasonality and their influences are provided in Supplementary Text 3.

Changes in growing season productivity and non-growing and shoulder season respiration determine SCA trends, with warming being a primary control on the seasonal dynamics of photosynthesis and respiration[33]. Plant photosynthesis and ecosystem respiration respond asynchronously to seasonality in environmental drivers, such that increased CO_2 fixation during the growing season can be offset by enhanced respiratory CO_2 loss in the shoulder and winter seasons. This seasonal compensation generally amplifies the SCA[34] (Fig. 4) and reflects the different sensitivities of above-ground and below-ground carbon cycle processes to environmental drivers, varying among biomes and environmental gradients in the NHL[26,35]. Climate warming

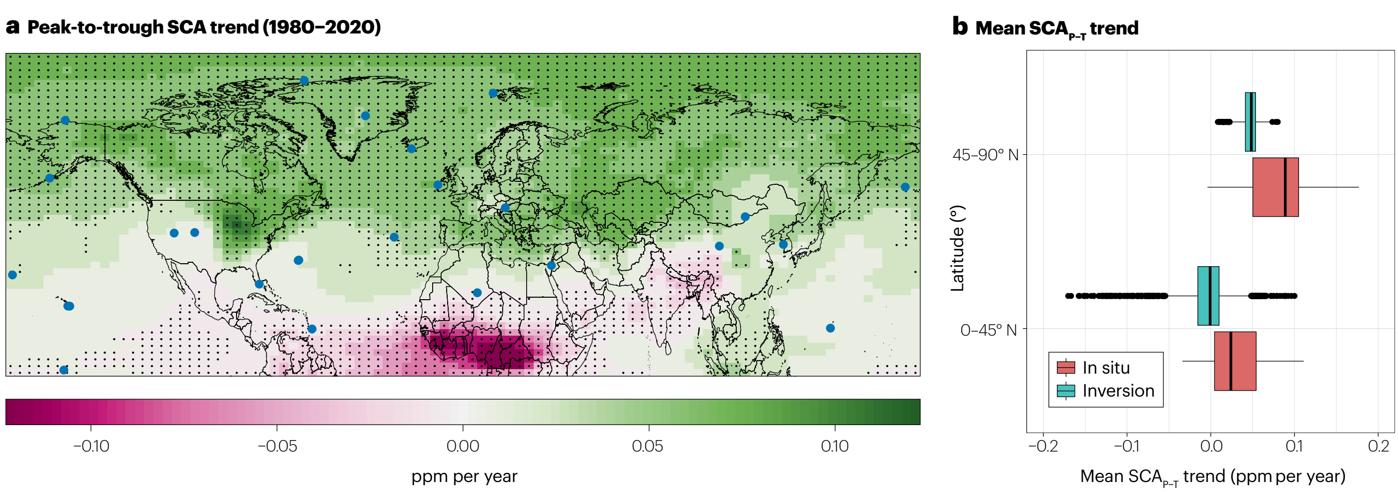


Fig. 2 | Observed trends in SCA$_{P\text{-}T}$. **a**, Trend in annual mean peak-to-trough seasonal cycle amplitude of atmospheric CO_2 (SCA$_{P\text{-}T}$) during 1980–2020 using inverse modelling data[178]; stippling indicates trends are significant at the 0.05 level. Blue dots represent in situ measurement sites. **b**, SCA$_{P\text{-}T}$ trends averaged over 0–45° N and 45–90° N for in situ[177] and inversion data[178]; boxes and whiskers represent interquartile range and outliers, respectively. SCA$_{P\text{-}T}$ increases are greater in the northern high latitudes (NHL) than at lower latitudes.

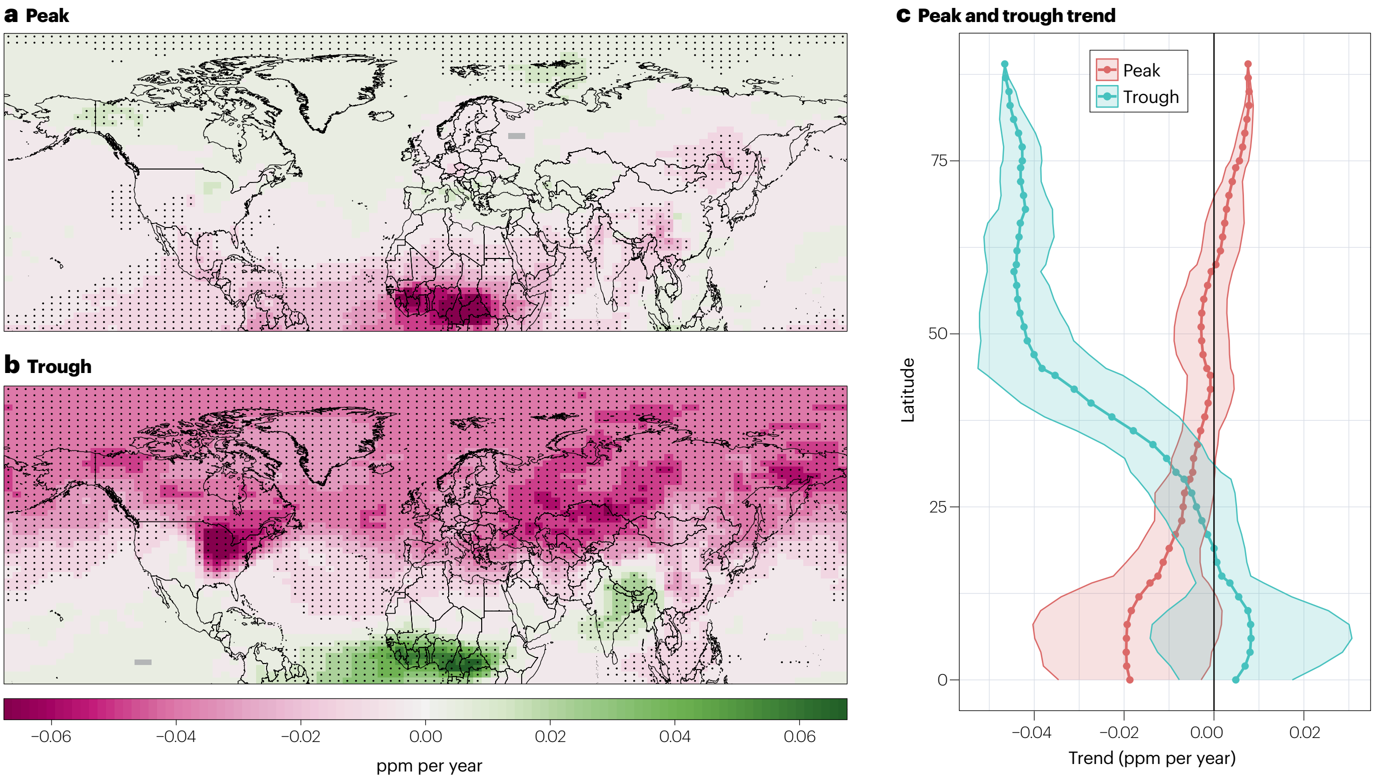


Fig. 3 | Observed peak and trough trends. a, Trend in the peak component of the seasonal cycle amplitude of atmospheric CO_2 (SCA) during 1980–2020 based on inverse modelling data[178]; stippling indicates trends significant at the 0.05 level. Trends represent a linear regression of the detrended peak (or trough) values against time (years). **b**, As in panel **a**, but for the trough component of the SCA. **c**, Latitudinal mean trends in peak (red) and trough (cyan) components of the SCA during 1980–2000; shading represents the standard deviation at each latitude. Enhanced growing season productivity is the major process contributing to increasing peak-to-trough seasonal cycle amplitude of atmospheric CO_2 (SCA_{P-T}) in the northern high latitudes (NHL).

also promotes earlier landscape thaw, reduced spring snow cover and earlier onset of vegetation productivity, and lengthens the growing season, overall favouring photosynthesis more than respiration[34]. Such warming-induced enhancement of CO_2 uptake by plants in spring and summer is the main mechanism explaining the increased seasonal CO_2 amplitude[8,36–38] (Figs. 5 and 6).

The season net CO_2 flux in autumn has contrasting responses to environmental drivers[11,39,40], making its potential contribution to increasing SCA unclear. Enhanced spring and summer productivity often increase cumulative evapotranspiration, reducing soil moisture and raising drought stress in the autumn[41–44]. Satellite observations over northern ecosystems confirm widespread moisture stress-induced declines in summer and autumn season productivity that offset productivity gains from warmer springs[33], although uncertainty in the spatial pattern and magnitude of these compensations remains. In addition, reduced light availability in the autumn season restricts photosynthesis in the Arctic, limiting any positive impact of warming on productivity[34,45]. Conversely, respiration in the autumn season can be either enhanced in response to warming[11], increased labile organic matter availability[46] and higher soil organic carbon turnover rates[47], or reduced due to soil moisture limitations. Thus, net CO_2 uptake during the autumn season can either increase[48] or decrease[49]. As the autumn season is an important driver for trends in the NHL SCA, there is a clear need to better understand key carbon cycle processes during this season.

Although heightened seasonality of terrestrial carbon fluxes is mainly driven by enhanced productivity, enhanced respiration is increasingly important to the changing SCA. Atmospheric CO_2 measurements[11,46,50] and site-scale eddy covariance data[12,26] show increased cold season respiration as a potentially important driver of the positive SCA trend. For example, during 1980–2020, net CO_2 uptake increased by 25% in spring and summer due to enhanced productivity, with a 15% decrease in autumn and winter attributed to enhanced respiration (Figs. 4b and 6; see Supplementary Table 2). But climate warming is also reducing snow cover area and duration[51], reducing thermal buffering and promoting colder winter soils. Although the response of winter respiration to climate is likely complex, owing to the interactive influence of snow conditions, air temperature and soil properties, observations indicate a net reduction in winter respiration across NHL ecosystems due to the influence of decreasing snow depth[52].

Overall, uncertainties associated with the NHL carbon cycle changes during autumn and winter lead to substantial uncertainties in the net CO_2 balance during those seasons, highlighting the importance of understanding seasonal climate–carbon feedback at high latitudes.

Biome and regional contributions

Spatial variability in the magnitude of SCA trends suggests distinct regional carbon cycle responses to climatic and environmental change and, thus, distinct contributions to the SCA[5,6,8] (Fig. 2a).

However, identifying a single dominant contributing biome to SCA change is challenging due to complex ecosystem–climate interactions and fast zonal atmospheric transport and mixing, complicating interpretation of the atmospheric amplification signals and identification of underlying mechanisms. Therefore, analyses typically refer to high-latitude Arctic and boreal ecosystems to denote broad regional contributions, rather than reporting on specific biomes. However, improved availability of in situ and remotely sensed observations and advanced modelling techniques are enabling better delineation of contributions from different biomes or ecoregions to the changing SCA[29].

Evidence suggests that boreal biomes are the dominant contributors to increasing SCA (Fig. 5). Boreal forests have stronger net CO_2 uptake, but increasing respiration during the autumn also enhances seasonal compensation of net CO_2 uptake (the offsets in net CO_2 uptake among seasons), resulting in boreal forests contributing to both the peak and the trough of the increase in SCA[26]. By comparison, tundra and temperate forests have a weaker net CO_2 uptake, resulting in a much smaller role in the increasing SCA in the NHL (Fig. 5).

The climatic and environmental characteristics of Arctic-boreal Eurasia and North America differ notably, influencing their relative contributions to SCA trends. Differences in regional climate characteristics and trends, permafrost distribution, vegetation composition and structure, wildfire regimes, land use histories and sea ice extent[53–55] suggest distinct regional responses to environmental drivers and varying contributions to increasing NHL SCA. Eurasia, especially Siberian larch forests, is increasingly recognized as a major regional contributor to increasing SCA[29]. Although these high-latitude systems are thought to be primarily temperature-limited, moisture availability during the peak of the growing season could also be important. There is evidence that moisture limitation in Eurasia has eased since the 2000s, either from enhanced atmospheric moisture transport due to an increasingly ice-free Arctic Ocean or from elevated water tables above continuous permafrost[56,57].

In contrast, boreal North America appears to be more drought sensitive, limiting seasonal CO_2 uptake. Satellite observations show widespread but non-uniform greening trends across the NHL regions, which are generally attributed to temperature and elevated CO_2 (ref. 58).

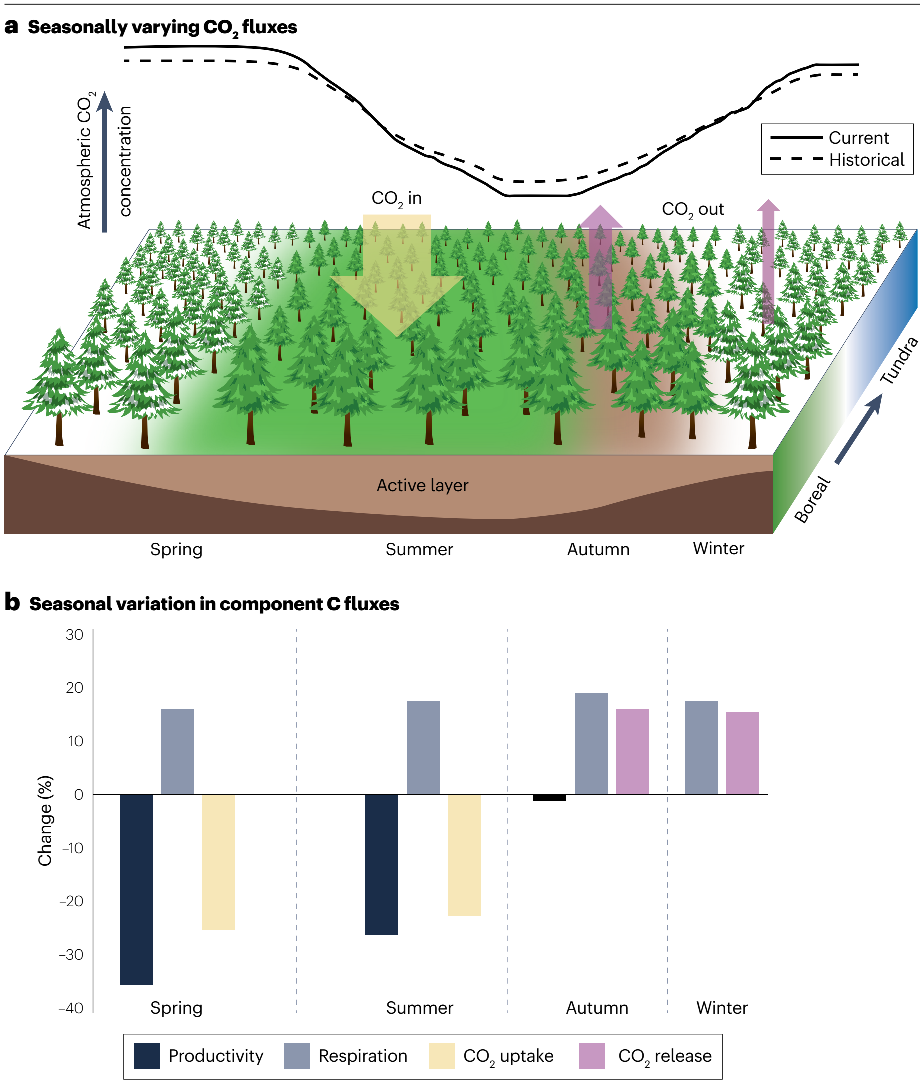


Fig. 4 | Seasonal variation in CO_2 fluxes across the northern high latitudes. a, Conceptual diagram of the seasonal changes in the carbon dioxide (CO_2) seasonal cycle (dashed and solid lines compare historical and current climate, respectively), the landscape and permafrost along a north–south thermal gradient. Yellow and purple arrows reflect CO_2 uptake and CO_2 release, respectively. **b**, Seasonal change in net CO_2 flux (yellow for net uptake from the atmosphere by land ecosystems, purple for net release from land ecosystems to the atmosphere) and its component carbon fluxes (black for productivity, grey for respiration) (data from Supplementary Table 2). Spring is defined as April–May; summer as June–August; autumn as September–October; and winter as November–March. Heightened seasonality in net terrestrial CO_2 exchange, due to enhanced growing season photosynthesis, is the main mechanism driving rising seasonal cycle amplitude of atmospheric CO_2 (SCA) in the northern high latitudes (NHL).

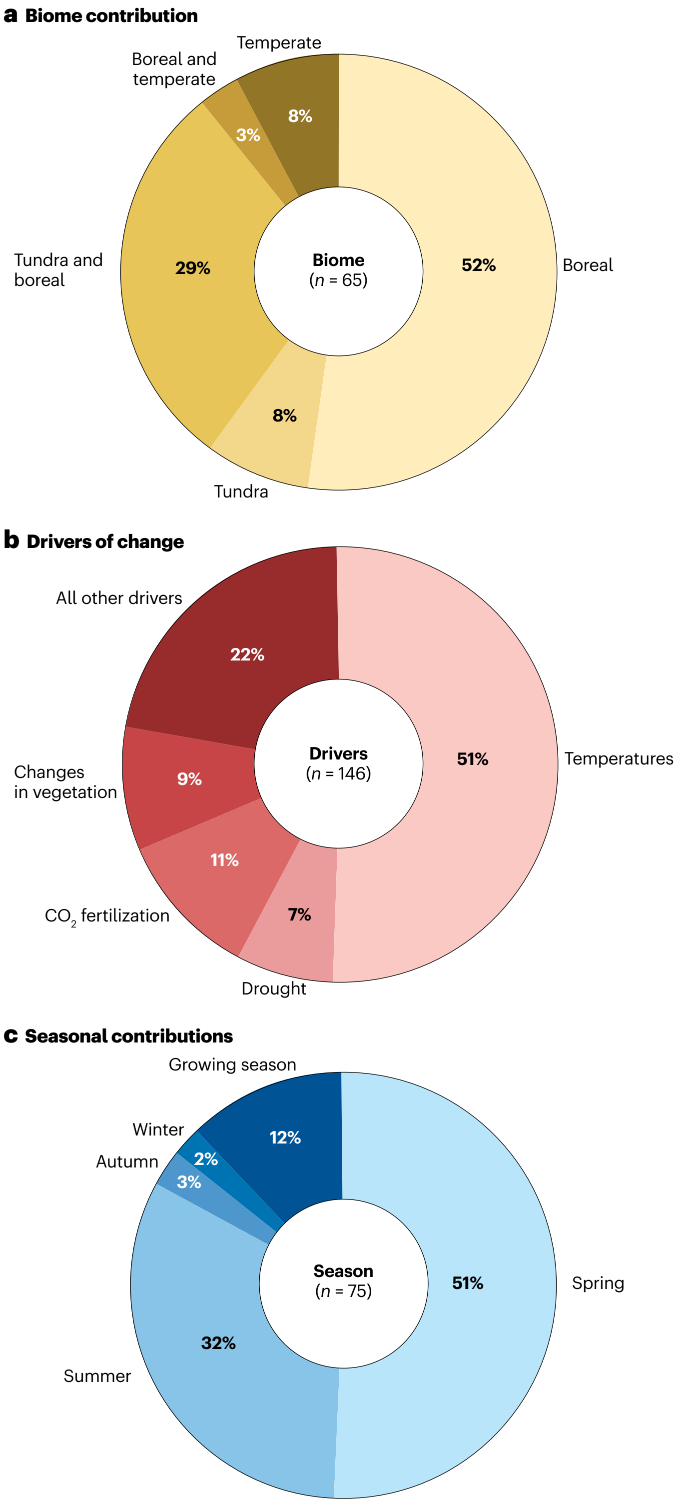


Fig. 5 | Relative influence of mechanisms driving increases in SCA. a, Biome contributions to changes in the seasonal cycle amplitude of atmospheric CO_2 (SCA) according to a literature review; *n* represents the number of published articles included in the literature review. **b**, As in panel **a**, but for the drivers of change. **c**, As in panel **a**, but for seasonal contributions (see Supplementary Figs. 2 and 3, and Supplementary Tables 2–4). Climate warming over the spring and summer season in Arctic and boreal ecosystems drives the enhanced seasonal cycle amplitude of carbon dioxide (CO_2) trends.

Greening in Arctic-boreal Eurasia has been more persistent than North American ecosystems[59–61], with greater increases in growing season length[40] matching regional patterns of net CO_2 uptake and contributions to the SCA[14,29]. Arctic-boreal North America shows more browning and higher sensitivities of productivity to moisture[43,62,63]. In North America, elevated moisture stress and climate warming have also increased tree mortality and wildfire activity, the majority of which are high-intensity stand-replacing fires with large carbon emissions and century-long regrowth trajectories, in contrast to Eurasia's dominant surface fire regime[64], which also complicates regional contributions to the SCA[65,66].

Overall, Eurasia has shown more pronounced positive coupling between temperature and vegetation productivity since the early 2000s than Arctic-boreal North America[67]. Atmospheric modelling[37] and records of enhanced CO_2 drawdown[68] also confirm that Eurasia contributes more to increasing SCA, owing to its enhanced productivity which is dominated by Siberian boreal forests and tundra. Conversely, impacts from Arctic-boreal North America are much smaller and geographically localized[29].

In addition to continental differences, latitudinal variability in the SCA trends could also be related to latitudinal gradients in the temperature sensitivities of the component carbon fluxes. Distinct latitudinal dependencies of the effects of warming on SCA trends reflect higher temperatures increasing SCA in high latitudes and decreasing SCA in mid-latitude regions, due to other drivers such as drought[17]. The effects of higher temperatures on productivity and respiration also differ among seasons, and therefore the effects of temperature changes on the SCA trends are seasonally and regionally dependent. Such effects are further complicated by interactions with seasonality in other environmental drivers, such as moisture availability, snow dynamics and permafrost[17].

Despite uncertainties, boreal biomes, especially Eurasia boreal forests, are identified as major contributors to increasing SCA in the NHL. Boreal regions within North America and Eurasia show distinct sensitivities to climate, potentially with several underlying mechanisms. However, observational data coverage across the Eurasian boreal region remains sparse, and is considerably lower than that of boreal North America, which contributes to the uncertainty in large-scale assessments of differential responses between the continents and the underlying drivers.

Drivers of seasonal carbon flux change

The seasonality of terrestrial CO_2 flux in the NHL is influenced by numerous interacting drivers operating at different spatial and temporal scales, varying seasonally, regionally and between biomes[69] (Fig. 6). Generally, regional to global drivers act as macro-scale drivers and provide boundary conditions for various ecosystem and environmental changes, whereas landscape-scale drivers, which are less well understood, change at a faster rate and are spatially more heterogeneous, often feeding back to macro-scale drivers. Warming temperature, CO_2 fertilization and drought are important regional to global-scale drivers, whereas change in vegetation, permafrost thaw, disturbance and snow dynamics are important landscape-scale drivers (Figs. 5 and 6). These factors and their relative influence on SCA trends are now discussed.

Regional to global-scale drivers

Regional to global-scale drivers are generally the first-order drivers of ecosystem change in the NHL. These drivers are typically characterized by comparatively gradual changes despite regional, seasonal and inter-annual variations. Regional to global-scale drivers can affect carbon cycling directly through their effects on plant physiology and indirectly through their influence on landscape-scale processes.

Warmer temperature. Warming temperatures are the most important driver of increasing seasonal terrestrial CO_2 fluxes. NHL ecosystems are primarily thermal energy-limited, thus temperature has a first-order control on the rates of ecosystem processes[70]. Arctic amplification — more rapid warming in higher latitudes[71–74] — results in stronger warming impacts on the component carbon fluxes in the NHL compared with other regions[75]. The contrasting seasonal effects of warming on productivity and respiration across seasons have a strong positive influence on increasing SCA (Fig. 6).

Temperature controls ecosystem productivity and respiration by directly influencing the metabolic rate of plants and microbes. Temperature also controls the duration of the NHL growing season[76]. Generally, photosynthetic capacity, such as the maximum rate of Rubisco activity

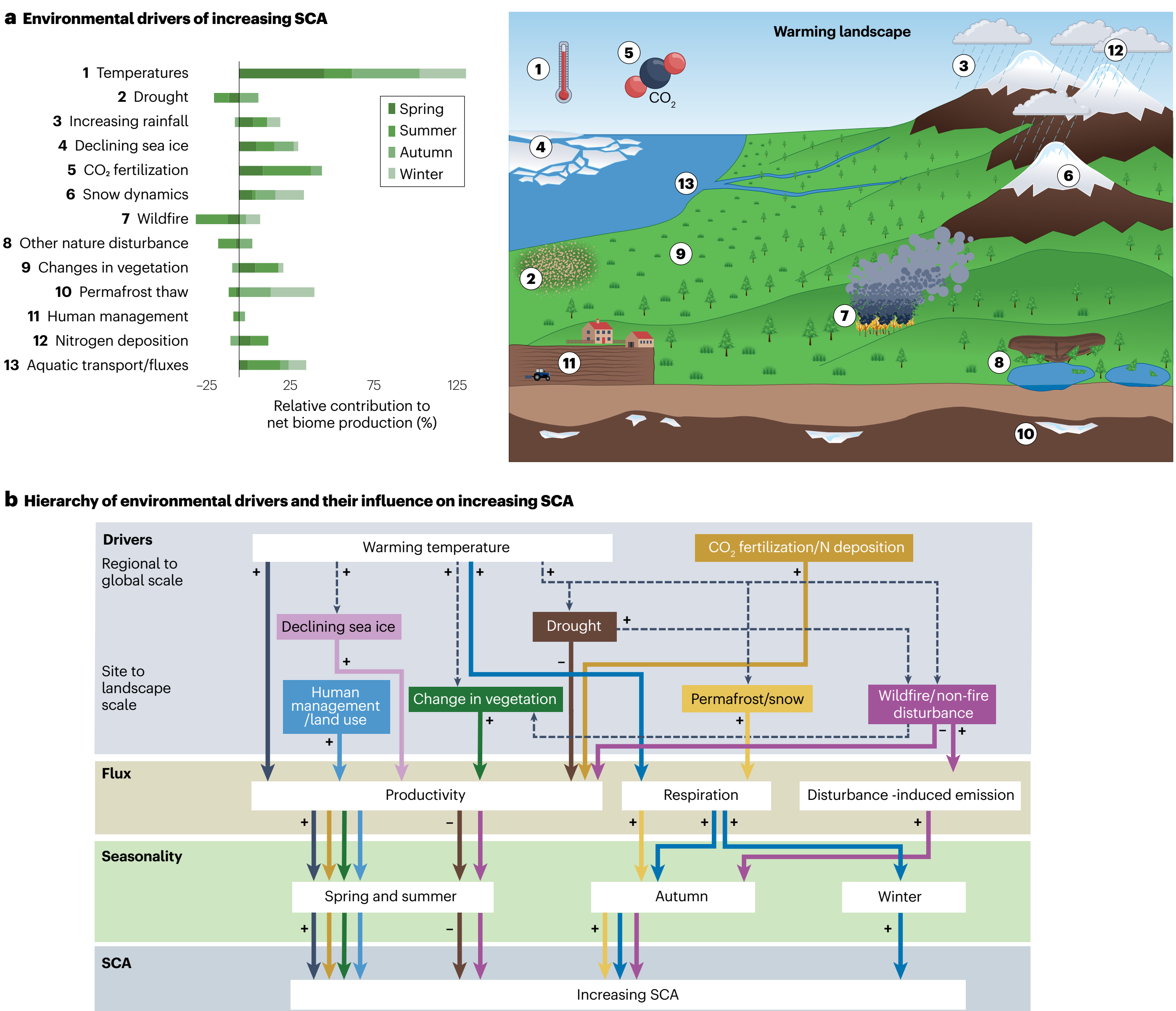


Fig. 6 | Environmental drivers influencing seasonal dynamics of SCA. **a**, Summary of expert opinion on the relative contribution of different drivers to net biome production (left) and the different drivers operating across a warming landscape (right). Drivers are numbered consistently between the graph and the diagram. **b**, Hierarchy of environmental drivers and their influence on increasing seasonal cycle amplitude of atmospheric CO_2 (SCA), informed by the expert survey (see Supplementary Tables 2–4). Colours correspond to specific drivers; signs (+ and –) correspond to driver influence (positive and negative, respectively) on the SCA. Changes in the seasonality of terrestrial carbon dioxide (CO_2) flux in the northern high latitudes (NHL) are influenced by numerous interacting drivers that operate at different spatial and temporal scales.

and the maximum rate of electron transport, increases with warming up to an optimal temperature in spring and summer[34], and then declines at higher temperatures due to enzyme denaturation, growth of metabolic respiration costs and other factors[77]. Most NHL ecosystems are below the optimum temperature for plant photosynthetic capacity[78,79], thus warming has increased productivity and Earth greening in the NHL[80–83]. However, remote-sensing observations suggest a weakening sensitivity of productivity to temperature due to other processes, such as vapour pressure deficit and biochemical reaction pathways[84] that negatively affect photosynthetic capacity as the temperature exceeds a certain threshold[79,85]. For example, as temperatures rise above the physiological optimum temperature of leaf-level photosynthetic capacity (typically 30 °C), stomatal closure to prevent excessive water loss reduces CO_2 uptake, and enzymatic efficiency declines due to biochemical limitations, resulting in decreased photosynthetic capacity[78].

Warming temperatures also increase respiration, as enzymatic activity and metabolic processes accelerate[86]. Respiration increases exponentially with warming[87], but evidence indicates that so far climate warming has enhanced productivity more than respiration, at least seasonally. Warming is a major driver of increased terrestrial net CO_2 uptake and SCA in the NHL[6,8,10], but the temperature sensitivities of productivity and respiration differ. Thus, future warming and associated changes in environmental drivers will likely impact component carbon fluxes differently, leading to a differential contribution to seasonal net CO_2 uptake and the SCA in the future[79].

Warmer temperatures can also have a lagged effect on seasonal water availability, which can change seasonal CO_2 dynamics and the net CO_2 balance[35]. For example, earlier snow-melt and stronger productivity in the early growing season can deplete soil water storage earlier, thus elevating drought stress in the late portion of the growing season[33]. Such increases in water stress could decrease productivity or increase respiration and compensate for the enhanced net CO_2 uptake in the early growing season[33,88,89], with seasonal compensation becoming larger with increasing tree cover and contributing to increasing SCA[26].

CO_2 fertilization and nitrogen deposition. Elevated atmospheric CO_2 concentration and enhanced nitrogen deposition can both have a positive influence on net CO_2 uptake. Elevated atmospheric CO_2 can increase leaf-scale photosynthesis and intrinsic water-use efficiency, resulting in higher plant growth, vegetation biomass, soil organic matter and carbon sinks[90]. Process modelling and free-air carbon enrichment experiments indicate that CO_2 fertilization has a strong positive effect on the SCA in the NHL, potentially being the second most important driver (Fig. 5). The effects of CO_2 fertilization on the SCA are mainly due to an increase in growing season productivity and, to a lesser degree, respiration (Fig. 6). Indeed, CO_2 fertilization is likely responsible for about half of the long-term increase in the terrestrial carbon sink[91]. However, the relative influence of CO_2 fertilization on the NHL SCA is highly uncertain[6,14], and the CO_2 fertilization effect on photosynthesis has been weakening since the 1980s, possibly due to emergent nutrient limitation by nitrogen and/or phosphorus[92].

Many NHL ecosystems are considered nitrogen-limited, such that nitrogen deposition contributes positively to net CO_2 uptake, and therefore the SCA, via productivity[93,94] (Fig. 6). Nitrogen limitation in plants primarily affects their growth and metabolic processes, leading to reduced chlorophyll production and photosynthesis, and slowing overall growth. Specifically, nitrogen scarcity limits a plant's ability to create essential organic compounds that are crucial for its growth and energy production. The main sources of nitrogen to the atmosphere are agricultural activities and fossil fuel burning, and, owing to short atmospheric lifetimes, the effects of nitrogen deposition on carbon cycle processes are regionally dependent and temporally variable[95]. Although CO_2 fertilization has likely influenced the SCA in the NHL, any corresponding influence from nitrogen deposition is likely more minor as the source regions for nitrogen are predominantly in the lower latitudes. Overall, the role of nitrogen deposition in shaping NHL SCA change is still poorly understood.

Drought. Drought is an important control on carbon cycle processes in NHL ecosystems. Potentially ranked fourth in contributing importance to the SCA based on the literature (Fig. 5), drought impacts the net water balance of ecosystems through atmospheric demand and soil water supply. Although increasing atmospheric temperatures have increased vapour pressure deficit, resulting in increased evaporative demand, soil moisture supply to plants is more complicated as it depends on precipitation, topographic characteristics, snow cover and permafrost conditions, with no clear regional trends since 2000 (refs. 96,97). In addition, soil moisture supply also affects carbon assimilation via root zone oxygen conditions. In cold permafrost regions, a prevalent wetting trend[98,99] is potentially transitioning to more widespread drying with continued permafrost thaw[100]. Drought stress could be intensifying in both tundra and boreal biomes, particularly in the summer and autumn seasons, altering the seasonality and magnitude of photosynthesis and respiration[101–103]. Increasing tree mortality, from drought and other factors, has also reduced boreal carbon sink capacity, affecting productivity, which could decrease the SCA[104] (Fig. 6).

Declining sea ice. Arctic sea ice decline has led to amplification of surface warming, resulting in an observed increase in plant productivity across the Arctic tundra[105], positively contributing to the SCA (Fig. 6). The relationships between trends of ice-free ocean extent and satellite-based vegetation indices are largely positive, but notable variations are found across the Arctic, and some regions even show strong local declines in productivity and greenness of above-ground tundra vegetation[106–108]. In addition, Arctic sea ice decline has the potential to increase the advection of moisture to the land, thus creating a complex impact on terrestrial carbon cycle processes, which could help explain different vegetation greening trends between Eurasia and North America[105].

Landscape to site-level drivers

Global environmental changes are reshaping the dynamics of NHL landscapes, impacting critical processes such as vegetation cover, land use, wildfires, permafrost thaw and snow dynamics. These landscape processes, which have key roles in regional carbon cycling and atmosphere CO_2 dynamics, are now discussed.

Change in vegetation composition and structure. Changes in vegetation composition and structure show regional and biome-specific patterns[109,110]. Climate warming and intensifying disturbance induce shifts in species composition and alter ecosystem structure and function[111–116]. Widespread woody encroachment along the tree–tundra ecotone and in the tundra of the NHL[36,117] has led to a shift in vegetation cover to plants that grow taller and thicker, contributing to higher net CO_2 uptake[118] and increased SCA (Fig. 6). Changes in vegetation are ranked as the third most important driver for increasing the NHL SCA (Fig. 5). With climate warming, tree species are also migrating northwards and to higher elevations at the leading edge of the boreal–tundra ecotone, with dieback at the trailing southern edge of their range[61,119,120].

Long-term remote sensing-based Earth observations have revealed a transition from the prevalent greening trend in lower tree cover regions to browning trends in higher tree cover regions, indicating changes in forest demography and function[119]. In boreal forests, greening trends are generally prevalent in cold and wet regions, with browning trends prevalent in relatively warm and dry regions[61]. Consequently, climate warming is most likely enhancing net CO_2 uptake in the tundra and in cooler and wetter portions of the boreal forest biome[121]. In the southern warmer and drier forest regions, increasing drought stress with higher risks of tree mortality and wildfire could lead to increases in net CO_2 release, due to decreased productivity and enhanced respiration, thereby decreasing the SCA[26,122].

Disturbance. Wildfires affect the carbon cycle through direct CO_2 emissions[123,124] and multiple post-fire carbon source and sink pathways[125], including changes in vegetation composition and structure, which are integral components of the global carbon cycle. Climate warming has increased the extent, frequency and severity of fire in boreal forests and Arctic tundra, and, as a result, has released large amounts of carbon through biomass and soil organic matter burning[126–129]. Fires occur predominantly in the summer and autumn seasons in the NHL, contributing positively to the SCA by increasing CO_2 emissions in late summer and early autumn. Conversely, more frequent and severe fires as well as biotic disturbances can reduce growing season productivity and CO_2 drawdown in recently burned stands, and therefore contribute negatively to the SCA (Fig. 6). Therefore, wildfires can either increase or decrease the SCA depending on the season in which they occur.

In addition, post-fire successional trajectories in forest composition and deepening of the active layer can influence the trends in the SCA. High severity and more frequent wildfires shift boreal forest composition towards younger stands with a greater proportion of deciduous species[130–133], which generally increases seasonal CO_2 exchange through shorter but stronger periods of CO_2 uptake[134–136]. Fires also deepen the active layer where younger forests have more access to nutrients enhancing CO_2 uptake[137]. The shortened period of CO_2 uptake can increase the influence of shoulder season respiration, which could also contribute to SCA enhancement. For example, site-level analysis from Canadian forests suggested that younger stands were associated with shorter growing seasons and stronger seasonality of net CO_2 uptake[138,139]. Therefore, changes in the seasonality of forest ecosystem fluxes following disturbance also affect atmospheric CO_2 concentrations and the SCA.

Permafrost thaw. Permafrost underlies ~25% of the northern hemisphere land surface and stores an estimated ~1,700 Pg (1,700 Gt) of carbon in frozen ground within the top 3 m[140,141]. Rapid anthropogenic warming and resulting thaw threaten to mobilize permafrost carbon stores. Permafrost degradation occurs gradually with active layer thickness increasing a few centimetres per decade, whereas abrupt thaw processes, such as thermokarst and thermoerosion, could expose many metres of permafrost carbon to mobilization on the timescale of days to a few years[142]. Both processes potentially increase atmospheric CO_2 concentrations and the SCA[143–146] (Fig. 6).

Another way permafrost thaw can change the SCA is through increasing productivity. Increased rooting depth of vegetation and nutrient supply, through enhanced groundwater and riverine transport, can lead to enhanced plant growth[147,148]. Therefore, permafrost thaw changes the thermal, hydrological and vegetation conditions that promote increasing below-ground carbon losses through respiration and/or increasing above-ground carbon gains through photosynthesis, both of which might enhance the SCA depending on their seasonal intensity[149–152].

Snow dynamics. Snow has an important role in regulating the surface energy balance, permafrost thaw and hydrology, plant phenology and terrestrial CO_2 cycles in NHL ecosystems[153]. Earlier seasonal snow-melt promotes longer growing seasons that alter the timing, magnitude and duration of plant photosynthesis and ecosystem respiration[154]. Such changes in earlier spring photosynthesis could be offset by late season declines in productivity due to drought[89] or by increases in early winter respiration after plants have senesced[46] that enhance the SCA, although the changes in annual net carbon balance might not be as strong as seasonal changes[155] (Fig. 6). Changes in snow cover can also have large impacts on ecosystems outside the growing season. Reductions in snow cover duration and depth increase the chance of spring frost damage, which can alter plant community composition, vegetation growth and carbon cycling[156–159].

Agricultural intensification and fossil fuel combustion. Increased agricultural productivity has enhanced net CO_2 uptake in cultivated lands, contributing to the increase in the SCA[18,28]. However, this occurs primarily in low to mid-latitude regions[160], and the estimated contribution to the overall increase in NHL SCA is likely small[6,10] (Fig. 6). Compared with other drivers, human land use and management play a minor role in influencing the SCA (see Supplementary Fig. 2).

Another human activity that directly influences the SCA is the combustion of fossil fuels. Fossil fuel emissions have increased from 2.6 Pg of carbon per year in 1960 to more than ~10 Pg of carbon per year in the 2020s (ref. 161). In mid and high-latitude regions, fossil fuel emissions are higher during winter months because of added demand for heating, which increases the peak CO_2 concentration during winter and early spring. However, fossil fuel emissions are expected to account for a small proportion of the NHL SCA trend[6,10].

Aquatic carbon transport. Aquatic ecosystems can contribute to changes in the SCA[162] through the seasonal transport of dissolved organic carbon (DOC) and exchange of CO_2 with the atmosphere. In particular, Arctic rivers constitute 10% of global discharge to the ocean[163] and export approximately 25–36 Tg of carbon per year[164]. Generally, pan-Arctic discharge has been increasing, with discharge rates accelerating more in Eurasian rivers than in North American rivers[165]. The discharge of DOC and nutrients is also variable, with most Arctic rivers showing a positive relationship between discharge and DOC concentration.

Furthermore, 50% of the DOC exported by Arctic rivers is relatively young (<5 years old) and labile. Therefore, increasing annual discharge and/or changes in DOC concentration could contribute to the increasing SCA, although DOC export is decreasing in some high-latitude rivers, such as the Yukon[166]. In contrast, dissolved inorganic carbon (DIC) and its contribution to CO_2 gas exchange render rivers a source of CO_2 to the atmosphere, with the greatest seasonal flux of CO_2 from arctic rivers to the atmosphere occurring in July[167]. Thus, transport and deposition of DOC is likely to enhance the SCA, but seasonal gas exchange of DIC is likely to dampen the SCA. Determining the role of NHL aquatic ecosystems in the Arctic carbon cycle requires more extensive and continuous measurements.

Relative importance of drivers

There is generally better data availability and understanding of the influence of warmer temperature, CO_2 fertilization and changes in vegetation on carbon fluxes compared with other drivers (Figs. 5 and 6a). Thus, there is greater confidence in the influence of these factors compared with other drivers (see Supplementary Table 3). However, expert opinion indicated large uncertainties but high importance of the impacts of additional drivers. These understudied drivers include changing snow cover and dynamics, changes in seasonal sea ice extent, permafrost thaw and changing disturbance regimes on seasonal changes in productivity and respiration, and their underlying mechanisms (Fig. 6a). Limited understanding of the impact of these landscape-scale drivers propagates into the net CO_2 balance of NHL ecosystems and seasonal interactions among the carbon cycle, climate and environmental conditions.

An atmospheric transport model prescribed by expert opinion quantified the potential contributions of unaccounted drivers to the SCA at Point Barrow (Box 1). At Point Barrow, temperature-driven trends within HNL ecosystems resulted in a 1.4 ppm increase in the SCA, arising from temperature impacts on net primary productivity (1.0 ppm). CO_2 fertilization also increased the amplitude of CO_2 by 1.4 ppm. However, unlike the case for primary production, increased heterotrophic respiration was largely not attributed to temperature and CO_2 fertilization but, rather, to a more complex set of drivers, including permafrost thaw, snowpack dynamics and wildfire (Box 1). Moreover, heterotrophic respiration is increasing faster than net primary production (NPP) in early winter, contributing substantially to increasing peak CO_2 concentration (Box 1).

Generally, the heightened seasonality of terrestrial carbon fluxes in the NHL is primarily driven by increased growing season plant productivity owing to warming and CO_2 fertilization. However, enhanced respiratory carbon release to the atmosphere during the non-growing season, which is an important factor with distinct drivers, underscores how changes in vegetation, snow, disturbance regimes and permafrost are key to capturing seasonal carbon cycle change in the NHL.

Summary and future perspectives

The increasing SCA in the NHL is one of the most evident fingerprints of global change on the biosphere, yet the underlying processes driving the seasonal patterns remain uncertain. Increasing seasonal CO_2 fluxes from arctic and boreal terrestrial ecosystems are the dominant contributor to this trend, likely driven by increased plant productivity during the early growing season and, to a lesser extent, enhanced respiration in the autumn and winter seasons. The contribution of warming and CO_2 fertilization are relatively well studied and understood. However, impacts from other important drivers, including changes in vegetation, drought, snow cover dynamics, heterogeneous sea ice decline, intensifying disturbance and permafrost thaw, are poorly constrained. Much of the variability in SCA trends is explained by increases in temperature and CO_2 fertilization, but other potentially important drivers, especially those regulating respiration and net CO_2 release in autumn and early winter, might change rapidly and deserve attention. Ecosystem-scale drivers might not yet be reflected in regional trends in the SCA, but as climate change progresses, their imprint on the SCA could be increasingly detectable.

The high spatial heterogeneity of CO_2 exchange and competing mechanisms complicates the interpretation of atmospheric data and the attribution of driving mechanisms. The interpretation and the attribution of SCA trends to drivers is further hindered by the relative scarcity of long-term CO_2 and ecosystem observations. In addition, current observational networks are biased towards relatively accessible regions and seasons. To address this issue, extended spatial coverage of in situ atmospheric and ecosystem observations to build a spatially and temporally representative monitoring network would help distinguish spatial gradients in the amplitude and phasing of CO_2 and elucidate underlying processes and mechanisms.

Prioritizing the expansion of existing CO_2 observational carbon eddy flux and concentration networks (such as FLUXNET and the circumpolar active layer monitoring network) into ecologically sensitive but under-represented biomes, such as tundra and Siberian larch forests, could help determine more accurate regional carbon balances[168–170]. Such network upgrades could capture rapid changes in land surface characteristics and support year-round measurements, providing insights into rapid changes and, also, shoulder and cold season dynamics. Complementary increases in observations and understanding of ecological changes, such as vegetation cover, permafrost dynamics, disturbance, and hydrological and thermal regimes, can provide valuable insight into how these factors interact to alter seasonal carbon dynamics[171]. These observations could be further supplemented with high spatio-temporal resolution remote-sensing observations that could provide insights into permafrost degradation, particularly if coupled with airborne and field measurements[172,173].

Another opportunity to better constrain SCA drivers and their influence is through improved simulation of the seasonal carbon cycle processes. Process models are one of the few tools that can decipher seasonal climate–carbon feedback in the Earth system, and project these into the future. However, state-of-art land surface models (LSMs) do not accurately simulate the SCA, partially because LSMs do not fully capture seasonal dynamics of net CO_2 fluxes or the influence of underlying drivers, which leads to large uncertainties[174–176]. For example, CO_2 fertilization and climate change are the most important drivers of increasing SCA, but there is large inter-model variability in the magnitude and direction of CO_2 sink-source dynamics resulting from these drivers[6,14]. In addition, LSMs generally underestimate net CO_2 release from respiration in the late-growing season and winter[12], thus underestimating the seasonal compensation of net CO_2 uptake and the SCA[26]. Therefore, LSMs need to better simulate the seasonal dynamics of photosynthesis and respiration and their drivers, particularly permafrost thaw, snow cover dynamics and wildfire. Such requirements reiterate the need for more long-term observations across a diversity of Arctic and boreal landscapes.

Understanding the mechanisms and drivers of the SCA is crucial for identifying and predicting large-scale environmental change and perturbations to the global carbon cycle. Therefore, understanding the multiple facets of the SCA requires more collaboration across many disciplines and among different research communities, including atmospheric scientists, ecologists, biogeochemists and climate scientists.

Data availability

All synthesized data are publicly available. The in situ carbon dioxide (CO_2) data are from the archive of the Earth System Research Laboratory, National Oceanic and Atmospheric Administration (NOAA)[177], and the Copernicus Atmospheric Modelling Service (CAMS) global inversion-optimized CO_2 concentration data are from the Copernicus Atmosphere Monitoring Service[178]. Code is available from the corresponding authors upon request.

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Author contributions

B.M.R. and Z.L. conceptualized the work. Z.L. led the work. Z.L., B.M.R., G.K.-A., Y.Z. and M.H. organized the work and contributed analysis. Z.L., B.M.R., G.K.-A., M.H., A.P.B., Y.Z. and J.S.K. wrote the original manuscript. All authors contributed to the discussion and writing of this manuscript.

Competing interests

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[1]Numerical Terradynamic Simulation Group, WA Franke College of Forestry and Conservation, University of Montana, Missoula, MT, USA. [2]Key Laboratory of Forest Ecology and Silviculture, Institute of Applied Ecology, Chinese Academy of Sciences, Shenyang, China. [3]Woodwell Climate Research Center, Falmouth, MA, USA. [4]Climate and Space Sciences and Engineering, University of Michigan, Ann Arbor, MI, USA. [5]Department of Physics & Atmospheric Science, Dalhousie University, Halifax, Nova Scotia, Canada. [6]GFZ German Research Centre for Geosciences, Potsdam, Germany. [7]Department of Ecosystem and Conservation Sciences, University of Montana, Missoula, MT, USA. [8]Laboratoire des Sciences des Climat et l'Environnement,

Gif-Sur-Yvette, France. [9]NASA Jet Propulsion Laboratory, California Institute of Technology, Pasadena, CA, USA. [10]National Center for Atmospheric Research, Boulder, CO, USA. [11]Cooperative Institute for Research in Environmental Sciences (CIRES), University of Colorado Boulder, Boulder, CO, USA. [12]NOAA Global Monitoring Laboratory, Boulder, CO, USA. [13]Center for Ecosystem Science and Society, Department of Biological Sciences, Northern Arizona University, Flagstaff, AZ, USA. [14]Department of Earth System Science, University of California Irvine, Irvine, CA, USA. [15]School of Biological Sciences, University of Utah, Salt Lake City, UT, USA. [16]Department of Earth, Atmospheric, and Planetary Sciences, Purdue University, West Lafayette, IN, USA. [17]GEODE Lab, School of Informatics, Computing, and Cyber Systems, Northern Arizona University, Flagstaff, AZ, USA. [18]Biological Science, University of Texas at El Paso, El Paso, TX, USA. [19]Department of Biology, Center for Ecosystem Science and Society, Northern Arizona University, Flagstaff, AZ, USA. [20]Joint Institute for Regional Earth System Science and Engineering, UCLA, Los Angeles, CA, USA. [21]Scripps Institution of Oceanography, La Jolla, CA, USA. [22]Department of Earth and Environmental Sciences, Columbia University, New York, NY, USA. [23]Sino-French Institute for Earth System Science, College of Urban and Environmental Sciences, Peking University, Beijing, China. [24]Northeast Institute of Geography and Agroecology, Chinese Academy of Sciences, Changchun, China. [25]Institute of Geography, University of Augsburg, Augsburg, Germany. [26]High Meadows Environmental Institute, Princeton University, Princeton, NJ, USA. [27]School of Life Sciences, Institute of Life Science and Green Development, Hebei University, Baoding, Hebei, China. [28]Key Laboratory of Ecosystem Network Observation and Modeling, Institute of Geographic Sciences and Natural Resources Research, Chinese Academy of Sciences, Beijing, China. [29]These authors contributed equally: Zhihua Liu, Brendan M. Rogers, Gretchen Keppel-Aleks, Manuel Helbig, Ashley P. Ballantyne, John S. Kimball.